

Fly GACA — PPL Mock Exam - Set A

SECTION 10-academy-curriculum

TYPE document

STATUS **active**

OWNER Founder

UPDATED 2026-06-14

Date: 2026-06-14

Level: Private Pilot Licence (PPL) — theoretical knowledge

Questions: 40

Time allowed: 50 minutes (recommended)

Pass mark: 75% (30/40)

Subject distribution: Reflects ICAO Annex 1 / GACA PPL knowledge areas

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SECTION 1 — AIR LAW & REGULATIONS (8 questions)

Q1. Which organisation is the civil aviation authority for the Kingdom of Saudi Arabia?

- A) The Ministry of Transport
- B) **The General Authority of Civil Aviation (GACA)**
- C) Saudi Arabian Airlines (Saudia)
- D) ICAO Regional Office, Cairo

Q2. GACAR Part 91 governs which category of operations?

- A) Air operator certification
- B) Pilot licensing requirements
- C) **General operating and flight rules**
- D) Aircraft airworthiness standards

Q3. When two aircraft are on a converging course at approximately the same altitude, which aircraft has the right of way?

- A) The faster aircraft
- B) The aircraft at the higher altitude
- C) **The aircraft to the other's right**
- D) The aircraft in controlled airspace

Q4. Under GACAR Part 91, what is the cloud clearance requirement for VFR flight below 10,000 ft in Class E airspace?

- A) Clear of cloud
- B) 500 ft below, 1,000 ft above, 1 SM horizontal
- C) **500 ft below, 1,000 ft above, 2,000 ft horizontal**
- D) 1,000 ft below, 1,000 ft above, 1 SM horizontal

Q5. An aircraft overtaking another aircraft must pass to which side of the aircraft being overtaken?

- A) Left (port) side
- B) Either side, at the pilot's discretion
- C) Above or below to avoid wake turbulence
- D) **Right (starboard) side**

Q6. What is the maximum indicated airspeed (IAS) permitted for operations below 10,000 ft MSL in controlled airspace under GACAR / ICAO standards?

- A) 200 kt
- B) 220 kt
- C) **250 kt**
- D) 300 kt

Q7. A pilot flying a VFR cross-country flight must obtain a weather briefing. Under GACAR Part 91, this pre-flight action is:

- A) Required only for international flights
- B) Required only when operating above 3,000 ft AGL
- C) **Required before any cross-country flight under IFR or VFR**
- D) Recommended but not required

Q8. Special VFR (SVFR) clearances in controlled airspace are issued by ATC to allow flight when weather is below standard VFR minima. Under GACAR, SVFR is typically authorised only:

- A) At night, automatically
- B) **When requested by the pilot and approved by ATC, with a minimum flight visibility of 1 SM**
- C) Whenever cloud base is below 1,000 ft
- D) To IFR-rated pilots only

SECTION 2 — METEOROLOGY (8 questions)

Q9. The International Standard Atmosphere (ISA) defines a sea-level temperature of:

- A) 10°C
- B) 20°C
- C) **15°C**
- D) 0°C

Q10. In the ISA, air temperature decreases with altitude at a standard lapse rate of approximately:

- A) 1°C per 1,000 ft
- B) 1.5°C per 1,000 ft
- C) **2°C per 1,000 ft**
- D) 3°C per 1,000 ft

Q11. Which cloud type is associated with severe turbulence, icing, lightning and hail and should be avoided by VFR pilots?

- A) Altostratus (As)
- B) Cirrus (Ci)
- C) Stratus (St)
- D) **Cumulonimbus (Cb)**

Q12. The QNH altimeter setting causes an altimeter to read:

- A) Height above a fixed datum of 1,013.25 hPa
- B) Height above the highest terrain on the route
- C) **Altitude above mean sea level based on the local station pressure reduced to sea level**
- D) Height above the aerodrome reference point

Q13. A METAR contains the group "BKN015". This reports:

- A) Broken cloud at 1,500 ft AMSL
- B) **Broken cloud at 1,500 ft AGL**
- C) Below-minimum cloud at 150 m
- D) Broken visibility in 15 km

Q14. What weather hazard is described as a sudden, violent downward and outward rush of air from a thunderstorm cell, causing severe horizontal wind-speed changes close to the ground?

- A) Thermal turbulence
- B) Mountain wave
- C) Clear-air turbulence (CAT)
- D) **Microburst / wind shear**

Q15. Carburettor icing is most likely to form under which atmospheric conditions?

- A) High temperature, low humidity
- B) Low temperature and low humidity
- C) **Temperature between -10°C and +25°C with high relative humidity**
- D) Any temperature above 0°C

Q16. During flight operations near the Saudi Red Sea coast and in the Hejaz, which regional weather hazard most frequently reduces visibility to below VFR minima and requires careful pre-flight meteorological assessment?

- A) Mountain wave turbulence
- B) Freezing rain and icing
- C) **Dust storms (haboob) and blowing dust (BLDU / BLSA in METAR)**
- D) Dense radiation fog

SECTION 3 — NAVIGATION (6 questions)

Q17. Lines running east and west parallel to the equator, used to measure angular distance north or south of the equator, are called:

- A) Meridians of longitude
- B) Isogonic lines
- C) Radials
- D) **Parallels of latitude**

Q18. What is the 1-in-60 rule used for in dead-reckoning navigation?

- A) Calculating fuel consumption at 60 kt
- B) Measuring compass deviation over 60 nm
- C) **Estimating the track correction angle when off-course: 1 nm off track in 60 nm flown equals 1° error**
- D) Setting transponder squawk codes

Q19. Magnetic variation (magnetic declination) is defined as the angular difference between:

- A) True north and compass north
- B) Magnetic north and compass north (compass deviation)
- C) **True north and magnetic north**
- D) Grid north and true north

Q20. A VOR radial is defined as:

- A) The bearing FROM the station as measured on a magnetic compass
- B) **The magnetic bearing FROM the VOR station to the aircraft**
- C) The bearing TO the VOR station from the aircraft
- D) The true bearing from the station

Q21. On an ICAO 1:500,000 aeronautical chart, 1 mm on the map represents how many metres on the ground?

- A) 50,000 m
- B) **500 m (since 1 mm = 1/500,000 of the ground distance, i.e., 500 m)**
- C) 5,000 m
- D) 50 m

Q22. A pilot departs OERK on a track of 270° (M) with a true airspeed (TAS) of 110 kt. The wind is from 360° at 20 kt. Using the wind triangle, the approximate wind correction angle (WCA) is:

- A) 0° — no crosswind component
- B) **Approximately 10° to the right (wind correction to port = left, so heading 280° to maintain 270° track)**
- C) 20° to the left
- D) 5° to the right

SECTION 4 — HUMAN PERFORMANCE & LIMITATIONS (5 questions)

Q23. Hypoxia resulting from reduced oxygen partial pressure at altitude first impairs which body system?

- A) The muscular system
- B) The respiratory system
- C) **The brain, affecting judgement, memory and coordination before the pilot is aware**
- D) Vision only

Q24. A pilot has taken an over-the-counter antihistamine for hay fever. What effect may this have?

- A) No effect — only prescription medicines affect flight safety
- B) Improved reaction time due to allergy relief
- C) **Possible drowsiness, impaired reaction time and degraded decision-making; the pilot should not fly**
- D) The effect is negligible above 3,000 ft

Q25. The "get-there-itis" hazardous attitude is most accurately described as:

- A) Refusing to follow ATC instructions
- B) Excessive confidence in personal skill
- C) **Pressing on to a destination despite deteriorating conditions or fuel concerns, overriding sound judgement**
- D) Distraction from non-flying tasks

Q26. Spatial disorientation is a particular hazard during which phase of flight?

- A) Cruise in VMC
- B) Take-off in calm conditions
- C) **Flight in IMC or at night without visual reference to the horizon**
- D) Landing in light crosswind

Q27. GACA medical certificates for pilots are governed by which GACAR Part?

- A) Part 43
- B) Part 61
- C) Part 91
- D) **Part 67**

SECTION 5 — PRINCIPLES OF FLIGHT (5 questions)

Q28. An aerodynamic stall occurs when:

- A) The aircraft's airspeed drops below 60 kt
- B) Engine power is reduced to idle
- C) **The wing exceeds its critical angle of attack, causing airflow to separate from the upper surface**
- D) The flaps are fully retracted

Q29. Which statement best describes induced drag?

- A) Drag caused by the friction of air over the airframe skin
- B) Drag caused by the undercarriage and external antenna
- C) Drag caused by exceeding critical Mach number
- D) **Drag that is a by-product of lift generation; it increases as angle of attack and lift increase**

Q30. What is the effect of a forward centre-of-gravity (CG) position on the stability of an aircraft?

- A) Increases the tendency to pitch up; reduces stability
- B) Has no effect on stability
- C) Reduces the stall speed
- D) **Increases longitudinal stability but requires more back-pressure to maintain level flight; increases stall speed**

Q31. The load factor in a 60° banked level turn is:

- A) 1.0 G
- B) 1.4 G
- C) **2.0 G**
- D) 2.5 G

Q32. Ground effect is most pronounced when the aircraft is flying at a height below:

- A) One wingspan above the surface
- B) **One wingspan or less above the surface**
- C) 500 ft AGL
- D) 50 ft AGL only during the landing flare

SECTION 6 — AIRCRAFT GENERAL KNOWLEDGE (4 questions)

Q33. During the pre-flight run-up, the magneto check is performed to:

- A) Test the fuel selector valve operation
- B) Verify the vacuum system pressure
- C) **Verify each magneto is functioning independently; a drop above the permitted limit indicates**

a fault

D) Check the propeller pitch mechanism

Q34. What is the purpose of the mixture control on a piston aircraft engine?

A) To control the propeller pitch

B) To regulate oil temperature

C) To select the fuel tank

D) **To adjust the fuel-to-air ratio for altitude and temperature, preventing over-rich or over-lean mixture**

Q35. A blocked pitot tube (with an open static port) will cause the airspeed indicator to:

A) Read zero at all airspeeds

B) Read correctly until above 10,000 ft

C) **Act like an altimeter — reading an airspeed that increases when climbing and decreases when descending**

D) Over-read by a fixed percentage

Q36. The squawk code 7700 is reserved for which condition?

A) Radio failure

B) Unlawful interference (hijack)

C) Parachute operations

D) **General emergency**

SECTION 7 — OPERATIONAL PROCEDURES (4 questions)

Q37. The standard Airmanship principle "Aviate, Navigate, Communicate" defines the priority order in an emergency. What does this mean in practice?

A) Always call ATC first, then maintain control

B) File a PIREP first, then navigate, then aviate

C) **Fly the aircraft first (maintain control); then establish or maintain position; then communicate with ATC**

D) Communicate with passengers first, then ATC

Q38. Wake turbulence produced by large aircraft is most hazardous to a following light aircraft in which situation?

A) When the light aircraft is directly above the heavy aircraft's flight path

B) In turbulent air, which disperses wake vortices quickly

C) **When landing behind a heavy aircraft in calm wind, following the same approach path at or below the heavy aircraft's approach path**

D) During cruise when separated by 5 nm laterally

Q39. The ATC light-gun signal of a flashing red directed at an aircraft on the ground means:

- A) Clear to take off
- B) Stop
- C) **Taxi clear of the runway/landing area in use**
- D) Return to starting point on the aerodrome

Q40. Before conducting a VFR cross-country flight, GACAR Part 91 requires the pilot in command to:

- A) File a defence notification with Saudi authorities
- B) Obtain a CAT II IFR clearance as backup
- C) Carry oxygen above 8,000 ft in all cases
- D) **Become familiar with all available information appropriate to the flight, including weather reports and forecasts, fuel requirements, alternatives, and NOTAMs**

ANSWER KEY

Q	Answer	Subject	GACAR / ICAO Reference
1	B	Air Law	Civil Aviation Law of KSA; GACAR Part 1
2	C	Air Law	GACAR Part 91
3	C	Air Law	GACAR Part 91, §91.181 (Rules of the Air) / ICAO Annex 2
4	C	Air Law	GACAR Part 91, §91.165 (VFR weather minima)
5	D	Air Law	GACAR Part 91, §91.183 (Overtaking) / ICAO Annex 2, §3.2.4
6	C	Air Law	GACAR Part 91, §91.117 / ICAO Annex 2
7	C	Air Law	GACAR Part 91, §91.103 (Pre-flight action)
8	B	Air Law	GACAR Part 91 SVFR provisions
9	C	Meteorology	ICAO Doc 7488 (ISA)
10	C	Meteorology	ICAO Doc 7488 — ISA DALR $\approx 2^{\circ}\text{C}/1,000\text{ ft}$ (1.98°C)
11	D	Meteorology	FAA Weather Handbook / ICAO Annex 3
12	C	Meteorology	ICAO Annex 3 / GACAR Part 91 altimetry
13	B	Meteorology	ICAO Annex 3, METAR standard (cloud height in AGL)
14	D	Meteorology	ICAO Doc 9817 — Low-Level Wind Shear
15	C	Meteorology	ICAO Aircraft Icing Manual / AFM guidance
16	C	Meteorology	Saudi AIP MET section; ICAO Annex 3 (BLDU/BLSA codes)
17	D	Navigation	ICAO Annex 4 / PHAK Chapter 15
18	C	Navigation	FAA PHAK / standard dead-reckoning technique
19	C	Navigation	ICAO Annex 4; standard navigation definitions

Q	Answer	Subject	GACAR / ICAO Reference
20	B	Navigation	ICAO Annex 10 / VOR ground station definition
21	B	Navigation	Map scale mathematics: $1 \text{ mm} \times 500,000 = 500,000 \text{ mm} = 500 \text{ m}$
22	B	Navigation	Wind triangle — southerly wind (from 360°) on a westerly track produces a northward drift; correct left to maintain track (head slightly north), approximately $\text{WCA} = \arcsin(20/110) \approx 10^\circ$
23	C	Human Performance	ICAO Doc 8984 (Human Factors Training) / FAA Hypoxia Brochure
24	C	Human Performance	GACAR Part 91, §91.17; GACA medical guidance
25	C	Human Performance	FAA Aeronautical Decision Making (ADM)
26	C	Human Performance	ICAO Doc 8984; FAA PHAK Chapter 17
27	D	Human Performance	GACAR Part 67
28	C	Principles of Flight	ICAO Annex 1 / standard aerodynamics
29	D	Principles of Flight	Standard aerodynamic theory
30	D	Principles of Flight	Standard aerodynamic theory / AFH
31	C	Principles of Flight	Load factor = $1/\cos(\text{bank angle})$; $\cos(60^\circ) = 0.5$; $\text{LF} = 2.0 \text{ G}$
32	B	Principles of Flight	Standard aerodynamic theory — ground effect below one wingspan
33	C	Aircraft General Knowledge	GACAR Part 43 / AFM run-up procedures
34	D	Aircraft General Knowledge	Standard piston engine theory / AFM
35	C	Aircraft General Knowledge	FAA PHAK Chapter 8 — pitot-static system malfunctions
36	D	Aircraft General Knowledge	GACAR Part 91, §91.285 / ICAO Annex 10 — transponder codes
37	C	Operational Procedures	Standard airmanship / ICAO Threat and Error Management
38	C	Operational Procedures	GACAR Part 91 / ICAO Doc 9426 (ATC Wake Turbulence)

Q	Answer	Subject	GACAR / ICAO Reference
39	C	Operational Procedures	GACAR Part 91, §91.141 / ICAO Annex 2 light signals table
40	D	Operational Procedures	GACAR Part 91, §91.103 (Pre-flight action)

Scoring Guide

Score	Result
30–40 (75–100%)	PASS — Proceed to scheduled ground examination preparation
25–29 (63–75%)	Borderline — Review weak subjects and resit
0–24 (below 63%)	FAIL — Additional study required across multiple subjects

Subject Breakdown (self-assess after marking)

Subject	Qs	Your Score	Max
Air Law & Regulations	1–8	/8	8
Meteorology	9–16	/8	8
Navigation	17–22	/6	6
Human Performance	23–27	/5	5
Principles of Flight	28–32	/5	5
Aircraft General Knowledge	33–36	/4	4
Operational Procedures	37–40	/4	4
Total	1–40	/40	40

Rationale Notes for Instructors / Self-Study

Q4 (VFR minima, Class E): GACAR Part 91 mirrors ICAO Annex 2 for Class E. The 500 ft / 1,000 ft / 2,000 ft values differ from Class G (below 1,200 ft AGL: 1 SM and clear of cloud). Verify current GACAR Part 91 §91.165 for the exact Saudi minima table.

Q16 (Saudi weather): This question targets Saudi-specific meteorological hazards not addressed in generic ICAO questions. Blowing dust (BLDU) and blowing sand (BLSA) are common METAR codes at inland Saudi stations and can reduce visibility to below 800 m with little warning. Check Saudi AIP MET for current observational reporting.

Q21 (Map scale): A 1:500,000 chart is the standard VFR en-route scale. 1 mm on paper = 500 m on ground = 0.27 nm approximately. 1 cm = 5 km = 2.7 nm.

Q22 (Wind triangle): A wind from 360° (true north) on a track of 270° creates a full 90° crosswind from the right (from the north, relative to a west-bound aircraft). The drift is to the south (left). To maintain a 270° track, the pilot turns left — a right wind correction angle. $WCA = \arcsin(V_w/TAS) = \arcsin(20/110) \approx 10^\circ$. Heading $\approx 260^\circ$ (M). *Verify with an E6B or aviation calculator.*

Q31 (Load factor 60° bank): $LF = 1/\cos(\theta)$. $\cos(60^\circ) = 0.5$. $LF = 2.0$ G. Stall speed in a 60° bank = $V_s \times \sqrt{2} \approx 1.41 \times$ straight-and-level V_s .

Q36 (Transponder codes): 7700 = Emergency; 7600 = Communication failure (lost comms); 7500 = Unlawful interference (hijack). These codes must be memorised; never select them inadvertently.

Q39 (Light-gun signals): For aircraft on the ground: flashing red = taxi clear of landing area. For aircraft in flight: flashing red = aerodrome unsafe, do not land. Steady red (ground) = stop; steady green (ground) = clear to take off.

End of Mock Exam Set A. For the next practice set, rotate subject ordering and use the Fly GACA quiz bank. Always verify every answer against the current official GACAR at gaca.gov.sa.